

I₂ molecules in superfluid He nanodroplets

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This is just a working document We address the zero temperature structure of superfluid helium nanodroplets doped with I₂ molecules within liquid ⁴He density functional theory (⁴He-DFT). The structure of helium nanodroplets doped with I atoms and I⁺ cations is also addressed. The number of He atoms in the first solvation shell is determined, as well as the solvation energy of these dopants in a ⁴He₂₀₀₀ nanodroplet. After exciting the molecule to the A state, the dynamics of I₂@He₂₀₀₀ is followed for some time within TDDFT. Subsequently, the I atoms are ionized and the evolution of the ions is followed for some time further.

I. INTRODUCTION

These notes are motivated by the exchanges we have had with Markus Koch and coworkers and aim to keep track of the progress we are making from the theory point of view. The final goal is to study the photodissociation of I₂ molecules from the A state of I₂ in He_N nanodroplets, as well as the dynamics of the I⁺ ions subsequently produced by the photoionization of I₂@⁴He_N.

Vibrational energy relaxation dynamics of I₂ molecules inside superfluid helium nanodroplets described within ⁴He-DFT has been studied by Vilà et al.¹ for a $N = 100$ ⁴He_N droplet. Photofragmentation of I₂ molecules inside helium clusters (droplets) has been studied by Panzou and Lewerenz² within the zero point average dynamics (ZPAD), which does not take into account the superfluid character of the droplet, using a quantum effective potential.

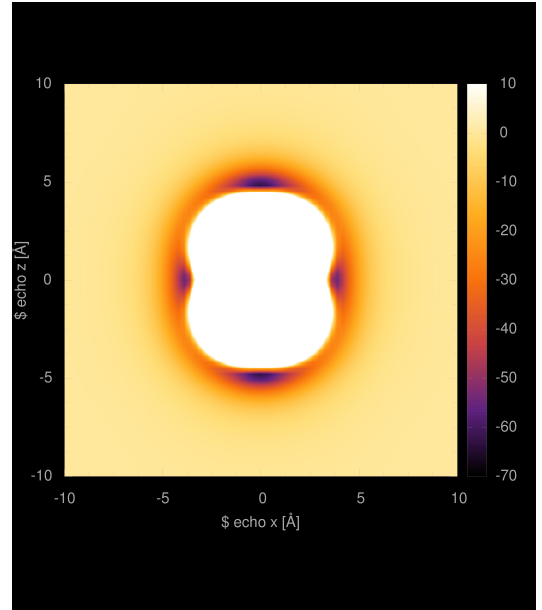


FIG. 1. I₂-He pair potential used in this work.⁸ Units are Å for lengths and K for the energy. The I₂ molecule lies along the z-axis.

II. THE HE-DFT APPROACH

Helium density functional and time-dependent density functional methods have proven to be very powerful tools to address superfluid ⁴He droplets. It is a phenomenological approach which constitutes a good compromise between accuracy and feasibility. The parameters of the functional have been adjusted to reproduce various properties of the bulk superfluid such as equilibrium density, energy per atom and compressibility, as well as the main features of the dispersion relation of the elementary excitations of superfluid ⁴He.³ A detailed description of the method can be found in Refs. 3–6. In this work we have treated the impurities as an external field in the statics and as classical particles in the dynamics.⁴

A. Statics

Within the DFT approach at zero temperature, the energy of a N -atom helium droplet ⁴He_N doped with a I₂ molecule, taken as an external field, is written as a functional of the ⁴He atom density $\rho(\mathbf{r})$ as

$$E[\Psi] = \int d\mathbf{r} \frac{\hbar^2}{2m_{\text{He}}} |\nabla \Psi|^2 + \int d\mathbf{r} \mathcal{E}_c(\rho) \quad (1)$$

$$+ \int d\mathbf{r} \mathcal{V}_{\text{He-I}_2}^{gs}(\mathbf{r}, \mathbf{r}_{\text{I}_2}) \rho(\mathbf{r}),$$

where the first term is the kinetic energy of the superfluid, m_{He} is the mass of the ⁴He atom and $\Psi(\mathbf{r})$ is the effective wave function (or order parameter) of the superfluid such that $\rho(\mathbf{r}) = |\Psi(\mathbf{r})|^2$ with $\int d\mathbf{r} |\Psi(\mathbf{r})|^2 = N$.

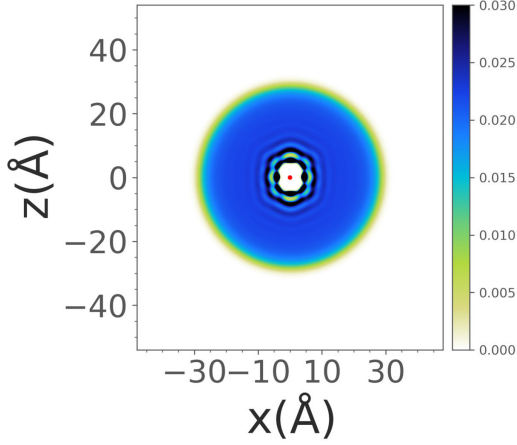


FIG. 2. Equilibrium configuration of $\text{I}_2@^4\text{He}_{2000}$: 2D atomic He density (in $\text{\AA}^{-3}$) on a symmetry plane containing the symmetry axis of the I_2 molecule, which lies along the z -axis (the red dot represents the I_2 center of mass). The color bar shows the density in units of $\text{\AA}^{-3}$.

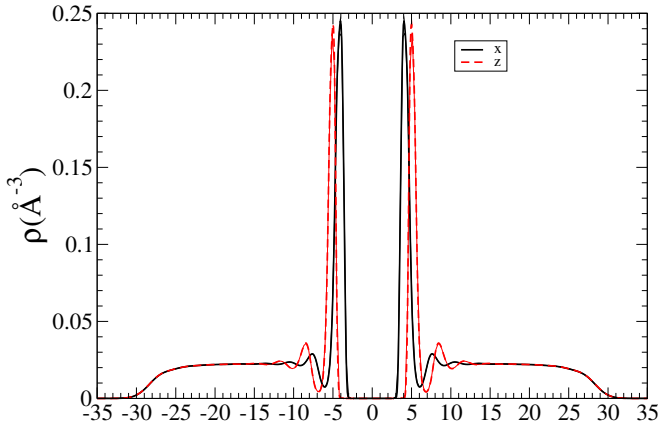


FIG. 3. Equilibrium configuration of $\text{I}_2@^4\text{He}_{2000}$: atomic He density (in $\text{\AA}^{-3}$) along the symmetry axis of the I_2 molecule (which lies on the z -axis) and perpendicularly to it (x -axis). Distances are in $\text{\AA}$.

The functional $\mathcal{E}_c(\rho)$ we have used contains the He-He interaction term within the Hartree approximation and additional terms describing non-local correlation effects.⁷ It is a modification of the Orsay-Trento functional³ which makes it stable even in the presence of very attractive impurities. The interaction of one single He atom with the I_2 molecule is represented by the potential $\mathcal{V}_{\text{He-I}_2}^{gs}(\mathbf{r}, \mathbf{r}_{\text{I}_2})$ ⁸ which depends on the I-I interdistance $|\mathbf{r}_{\text{I}_2}| = |\mathbf{r}_1 - \mathbf{r}_2|$, on the distance of the He atom to the center of mass of the I_2 $|\mathbf{r}|$, and on the angle θ between the vectors $\mathbf{r}$ and $\mathbf{r}_{\text{I}_2}$.

The droplet equilibrium configuration is obtained by solving the Euler-Lagrange (EL) equation arising from

functional variation of Eq. (1)

$$\left\{ -\frac{\hbar^2}{2m_{\text{He}}} \nabla^2 + \frac{\delta \mathcal{E}_c}{\delta \rho} + \mathcal{V}_{\text{He-I}_2}^{gs}(\mathbf{r}, \mathbf{r}_{\text{I}_2}) \right\} \Psi \equiv \mathcal{H}[\rho] \Psi = \mu \Psi, \quad (2)$$

where μ is the ^4He chemical potential corresponding to the number of He atoms in the droplet ($N = 2000$ in this work, corresponding to a droplet of $R = 28 \text{ \AA}$ radius⁵), and $\mathcal{H}$ is the DFT Hamiltonian

$$\mathcal{H} = -\frac{\hbar^2}{2m_{\text{He}}} \nabla^2 + \frac{\delta \mathcal{E}_c}{\delta \rho(\mathbf{r})} + \mathcal{V}_{\text{He-I}_2}^{gs}(\mathbf{r}, \mathbf{r}_{\text{I}_2}) \quad (3)$$

The EL equation has been solved by a relaxation (imaginary time) method using the ^4He -DFT-BCN-TLS computing package,⁹ see Refs. 4 and 6 and references therein for additional details. We work in Cartesian coordinates, with the effective wave function $\Psi(\mathbf{r}, t)$ defined at the nodes of a 3D grid inside a calculation box discretized using a $0.3 \text{ \AA}$ space step. Periodic boundary conditions (PBC) are imposed which allow to use the Fast Fourier Transform¹⁰ to efficiently compute the convolutions needed to obtain the DFT mean field $\mathcal{H}[\rho]$. The differential operators in $\mathcal{H}[\rho]$ are approximated by 13-point formulas.

The static calculations allow to obtain the solvation energy S_{I_2} of the I_2 molecule into a $^4\text{He}_N$ droplet. In general, the solvation energy of an impurity Y is defined as

$$S_Y = E(Y@^4\text{He}_N) - E(^4\text{He}_N). \quad (4)$$

These energies depend on N . For the sake of completeness, we will also calculate S_{I} and S_{I^+} .

B. Dynamics

In a first step, the iodine molecule is excited to the dissociative A-state (pump process). Next, it is photoionized, $\text{I} + \text{I} \rightarrow \text{I}^+ + \text{I}^+$ (probe process). Other probe processes, such as $\text{I} + \text{I} \rightarrow \text{I} + \text{I}^+$, are not considered in the theoretical description.

After determining the equilibrium configuration from a static calculation with $|\mathbf{r}_{\text{I}_2}| = 2.666 \text{ \AA}$, the dynamics is triggered substituting the ground state He- I_2 interaction by that corresponding to the A state, $\mathcal{V}_{\text{He-I}_2}^A(\mathbf{r}, \{\mathbf{r}_i\})$.

Treating the I_2 molecule as a classical particle, the energy of the system in the pump process is written as

$$\begin{aligned} E[\Psi, \{\mathbf{r}_i\}] &= \int d\mathbf{r} \frac{\hbar^2}{2m_{\text{He}}} |\nabla \Psi|^2 + \int d\mathbf{r} \mathcal{E}_c(\rho) \\ &+ \int d\mathbf{r} \rho(\mathbf{r}) \mathcal{V}_{\text{He-I}_2}^A(\mathbf{r}, \{\mathbf{r}_i\}) \\ &+ \sum_{i=1,2} \frac{1}{2} m_{\text{I}} \dot{\mathbf{r}}_i^2 + \mathcal{V}_{\text{I}_2}^A(|\mathbf{r}_{12}|) \end{aligned} \quad (5)$$

where the I-I interaction in the A state $\mathcal{V}_{\text{I}_2}^A(|\mathbf{r}_{12}|)$ is now explicitly included since the iodine atoms will participate in the dynamics.

The interaction potential between a He atom and the I₂ molecule $\mathcal{V}_{\text{He-I}_2}^A(\mathbf{r}, \{\mathbf{r}_i\})$ has been calculated in the first-order intermolecular diatomics-in-molecule perturbation theory approach (IDIM-PT1) of Buchachenko and Stepanov,^{11,12} which yields¹²

$$\mathcal{V}_{\text{He-I}_2}^A(\mathbf{r}, \{\mathbf{r}_i\}) = \mathcal{V}_{\text{He-I}}^A(|\mathbf{r} - \mathbf{r}_1|, \theta_1) + \mathcal{V}_{\text{He-I}}^A(|\mathbf{r} - \mathbf{r}_2|, \theta_2), \quad (6)$$

where

$$\begin{aligned} \mathcal{V}_{\text{He-I}}^A(|\mathbf{r} - \mathbf{r}_i|, \theta_i) = & \frac{1}{2} \left\{ \frac{3}{2} V_{\Pi} + \frac{1}{2} V_{\Sigma} - \frac{1}{2} (V_{\Pi} - V_{\Sigma}) \cos^2 \theta_i \right. \\ & \left. - \frac{1}{\sqrt{6}} (V_{\Pi} - V_{\Sigma}) \sin^2 \theta_i \right\} = \\ & \frac{1}{2} \left\{ \frac{3}{2} V_{\Pi} + \frac{1}{2} V_{\Sigma} - \frac{1}{\sqrt{6}} (V_{\Pi} - V_{\Sigma}) \right\} + \\ & \frac{1}{2} \left\{ \left(\frac{1}{\sqrt{6}} - \frac{1}{2} \right) (V_{\Pi} - V_{\Sigma}) \right\} \cos^2 \theta_i, \end{aligned} \quad (7)$$

where the argument $|\mathbf{r} - \mathbf{r}_i|$ of the V_{Σ} and V_{Π} potentials, taken from Ref. 13, has been omitted for clarity.¹⁴ The cosine-independent and cosine-dependent terms of the $\mathcal{V}_{\text{He-I}}^A(|\mathbf{r} - \mathbf{r}_i|, \theta_i)$ potential are plotted in Fig. 4. The cosine-independent component presents a minimum of -18.0 cm^{-1} at $|\mathbf{r} - \mathbf{r}_i| = 4.1 \text{ \AA}$, and it is the *dominant* part of the $\mathcal{V}_{\text{He-I}}^A(|\mathbf{r} - \mathbf{r}_i|, \theta_i)$ potential, diverging faster than the cosine-dependent part at distances shorter than their crossing point.

The I-I interaction potential $\mathcal{V}_{\text{I}_2}^A(|\mathbf{r}_{12}|)$ has been taken from Ref. 15, and is plotted in Fig. 5. Also plotted is its r -derivative which enters Eq. (9); it has a small structure between $r = 5.5 - 5.75 \text{ \AA}$ that hardly will have an effect on the dynamics.

The time evolution equations of the system are obtained by minimizing the action¹⁶

$$\begin{aligned} \mathcal{A}[\Psi, \{\mathbf{r}_i\}] = & \int dt \left\{ i\hbar \int d\mathbf{r} \Psi^*(\mathbf{r}, t) \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) \right. \\ & \left. + \sum_i m_i \dot{\mathbf{r}}_i^2 - E[\Psi, \{\mathbf{r}_i\}] \right\} \end{aligned} \quad (8)$$

Variation of $\mathcal{A}$ with respect to Ψ^* and $\mathbf{r}_i$ yields

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \Psi = & \left[-\frac{\hbar^2}{2m_{\text{He}}} \nabla^2 + \frac{\delta \mathcal{E}_c}{\delta \rho(\mathbf{r})} + \mathcal{V}_{\text{He-I}_2}^A(\mathbf{r}, \{\mathbf{r}_i\}) \right] \Psi \\ m_i \ddot{\mathbf{r}}_i = & - \left\{ \int d\mathbf{r} \mathcal{V}_{\text{He-I}}^A(|\mathbf{r} - \mathbf{r}_i|, \theta_i) \nabla \rho(\mathbf{r}) \right. \\ & \left. + \left[\frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}} \frac{d}{dr} \mathcal{V}_{\text{I}_2}^A(r) \right]_{r=r_{ij}} \right\} \quad (i = 1, 2, j \neq i), \end{aligned} \quad (9)$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ and the time dependence of the variables has been omitted for clarity. In order to solve Eqs. (9), initial values for the $\{\mathbf{r}_i, \dot{\mathbf{r}}_i\}$ variables and the effective wave function $\Psi(\mathbf{r})$ have to be specified. In our case, they will be the effective wave function of the droplet

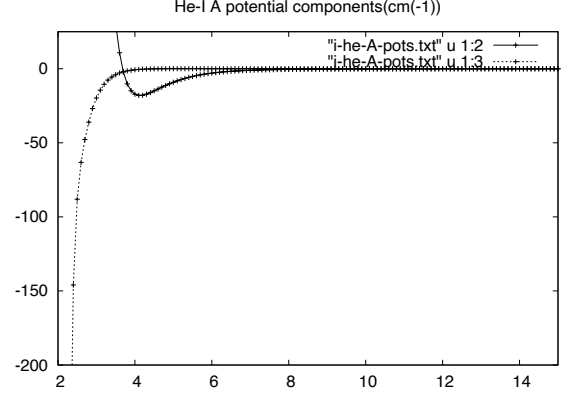


FIG. 4. Cosine-independent and cosine-dependent components of the $\mathcal{V}_{\text{He-I}}^A(|\mathbf{r} - \mathbf{r}_i|, \theta_i)$ potential.

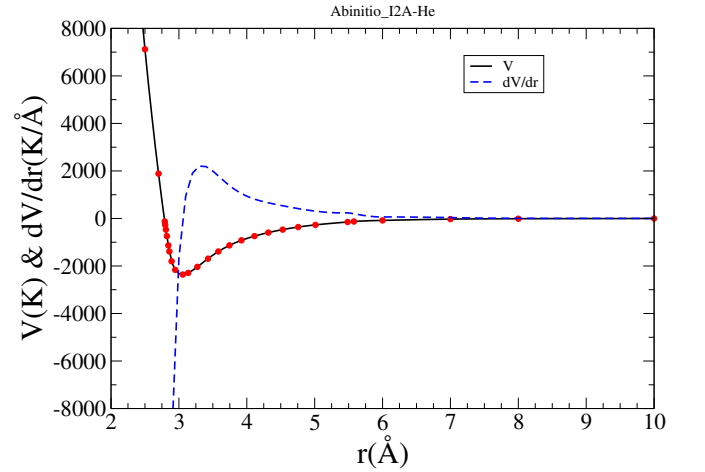


FIG. 5. I-I A *ab initio* interaction potential (K) as a function of r ($\text{\AA}$) (dots) connected by a cubic spline. Also drawn is its radial derivative.¹⁵

ground state hosting the I₂ molecule, with the iodine atoms set at the equilibrium distance $r_{12} = 2.666 \text{ \AA}$ and zero velocity. It can also be the static configuration corresponding to a constrained calculation, see below.

Equations (9) have been solved as indicated in Ref. 6 by using Hamming's predictor-modifier-corrector method¹⁷ initiated by a fourth-order Runge-Kutta-Gill algorithm.¹⁷ The electronic excitation of the I₂ molecule yields high energy excitations in the ⁴He droplet. Therefore, a small time-step (??? fs) is required during the first stages of the simulation, which makes the simulation very time-consuming. After a few ps, a larger time-step (??? fs) has been used.

Besides prompt emitted He atoms produced by the electronic excitation of the molecule, the very energetic excitations caused in the droplet (phonons, rotons, ripplons, ...) eventually decay by He atom ejection. These

He atoms reach the simulation box boundaries and re-enter the box from the opposite side due to the imposed PBC, interfering with the droplet in an unphysical and unpredictable way and leading to gross errors in the calculations. To avoid this, a buffer inside the calculation box is defined where an absorption potential acts to prevent the evaporated He atoms from re-entering.¹⁸

III. RESULTS

A. Statics

1. $I_2@He_{2000}$

We have used the I_2 -He pair potential of García-Gutierrez et al.⁸ The appearance of this potential on a plane containing the symmetry (z) axis of the I_2 molecule is shown in Fig. 1. For the equilibrium I-I bond distance $r_e = 2.666$ Å, the potential displays a global minimum for a linear configuration at a distance between the He atom and the center of mass of the I_2 molecule of $R_{He-I_2} = 4.828$ Å; it has an energy of -44.28 cm⁻¹. A second minimum, corresponding to a T-shaped configuration, appears at $R_{He-I_2} = 3.818$ Å with an energy of -38.92 cm⁻¹.⁸ The linear and T-shaped minima are clearly visible in Fig. 1. This topology is specific of dihalogens, not being present, e.g., in di-alkalis. Figure 2 shows the equilibrium configuration of $I_2@He_{2000}$ in a symmetry plane containing the symmetry axis of the I_2 molecule. The total energy, Eq. (1), is -12090.43 K (Ernesto has got -12089.50 K because we are using different interpolation schemes of the *ab initio* points of Ref. 8).

We note that the density close to the I_2 molecule is characterized by a well-defined inner (solvation) shell of He atoms around it. The average density in the first shell is well-above the freezing density of bulk liquid 4He (0.026 Å⁻³). Although this might be considered as an indication of a He solid-like structure in the first solvation shell, to clearly assess the solid-like or liquid-like character of the structure around the cation, one should also consider the dynamical behavior of the helium atoms. This is beyond the capabilities of He-DFT theory, but not of other more fundamental methods as the path integral Monte Carlo (PIMC) approach.^{19–21}

Figure 3 shows the atomic He density (in Å⁻³) along the symmetry axis of the I_2 molecule (which lies on the z -axis) and perpendicularly to it (x -axis). Distances are in Å. This figure compares well with Fig. 1 of Ref. 1, remember that the later is for $N = 100$.

The number of helium atoms around the I_2 molecule can be estimated carrying out a spherical average of the helium density around the center of mass of the molecule and integrating it from zero to the first minimum of the density profile. **THIS IS FAR FROM OBVIOUS**

We have calculated the solvation energy of I_2 at the center of the $^4He_{2000}$ droplet (equilibrium configuration)

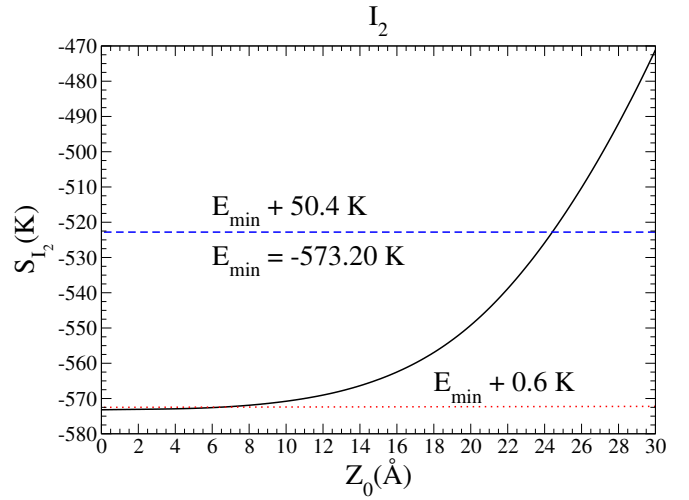


FIG. 6. Solvation energy of $I_2@He_{2000}$ as a function of the distance to the equilibrium position at the droplet center. The molecule is radially oriented and the energy range goes from -580 to -510 K. The horizontal lines represent the thermal $(3/2)k_B T$ energy (bottom line) and Landau $(1/2)m_{I_2}v_L^2$ energy (top line).

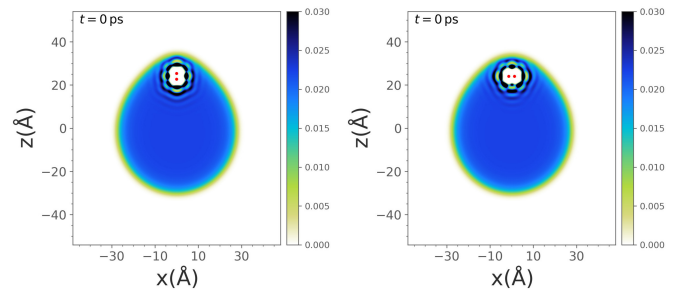


FIG. 7. Two-dimensional density plots of $I_2@He_{2000}$ with the I_2 molecule is at a distance of 24 Å from the center of the droplet. Left panel, radial orientation. Right panel tangential orientation. The color bar shows the density in units of Å⁻³.

and have obtained $S_{I_2} = -573.3$ K.

It is also useful to have the structure of $I_2@He_{2000}$ at locations different from the equilibrium one at $\mathbf{r} = 0$ and for different orientations. This can be done by constrained calculations.^{22–24} Figure 6 shows the solvation energy of $I_2@He_{2000}$ as a function of z . The molecule is radially oriented. The horizontal lines represent the thermal $(3/2)k_B T$ and Landau $(1/2)m_{I_2}v_L^2$ energies. Two dimensional plots of the He density corresponding to $I_2@He_{2000}$ with the I_2 radially and tangentially oriented at a distance of 24 Å, close to the classical turning point of I_2 for the Landau velocity, are presented in Fig. 7. The solvation energy when I_2 is at 24 Å radially oriented is -525.78 K, and -529.08 K when it is tangentially oriented (compare with the value of -573.19 K when I_2 is at the center of the droplet).

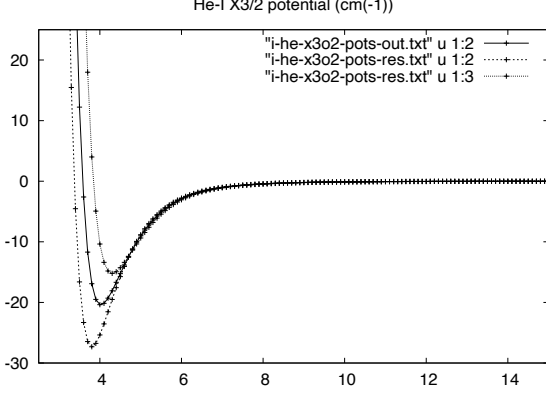


FIG. 8. From bottom to top, V_{Σ} , $V_{X,3/2}$ and V_{Π} He-I potentials¹³ (cm^{-1}) as a function of r ($\text{\AA}$). The equilibrium points are at (3.80, -27.32), (4.03, -20.40) and (4.30, -15.21) ($\text{\AA}$, cm^{-1}), respectively.

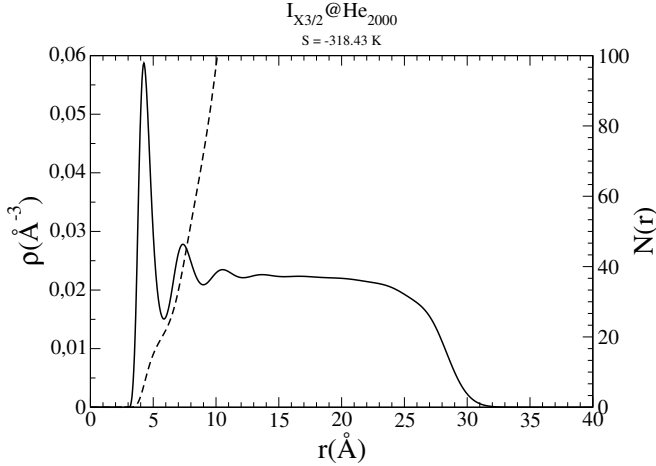


FIG. 9. Solid line, left scale: Helium atomic density profile corresponding to $\text{I@}^4\text{He}_{2000}$ when I is in the $X3/2$ state. Dashed line, right scale. Number of helium atoms as a function of the distance from the I atom.

2. I@He_{2000}

Buchachenko et al. have calculated the He-I $2\Sigma^+$ and 2Π pair potentials.¹³ Using them, we have addressed the structure of a $^4\text{He}_{2000}$ droplet doped with a I atom in the ground state taken into account the strong spin-orbit of iodine. The ground state of iodine has $J = 3/2$, and its interaction with a He atom is¹³

$$V_{X,3/2} = \frac{1}{2}(V_{\Sigma} + V_{\Pi} + \Delta_{SO} - \delta), \quad (10)$$

where $\Delta_{SO} = 7603.15 \text{ cm}^{-1}$ and

$$\delta = [(V_{\Sigma} - V_{\Pi})^2 - 2(V_{\Sigma} - V_{\Pi})\Delta_{SO}/3 + \Delta_{SO}^2]^{1/2}.$$

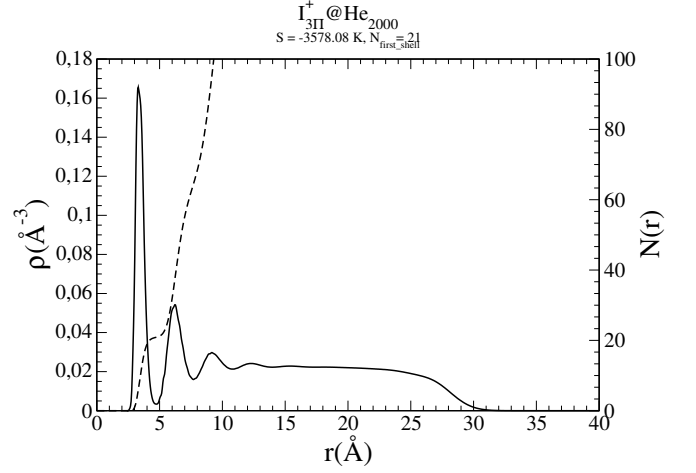


FIG. 10. Spherically averaged helium atomic density profile corresponding to $\text{I}^+@^4\text{He}_{2000}$ when I^+ is in the 3Π state.

The $V_{X,3/2}$, V_{Σ} and V_{Π} potentials are plotted in Fig. 8.¹⁴

Due to the large He-I interaction compared to the He-He interaction, the iodine atom resides in the bulk of the droplet. Its equilibrium position is at the center of the droplet, with a fairly large delocalization due to its zero point motion,² which is a general situation in doped helium droplets, see e.g. Ref. 22 for the case of Mg atoms.

The helium density profile of $\text{I@}^4\text{He}_{2000}$ at equilibrium, obtained when iodine is in the $X3/2$ state, is shown in Fig. 9. It can be seen that the effect of the impurity on the He droplet, even if it is sizeable, is local. The solvation energy of He in this state is $S_{IX3/2} = -318.43 \text{ K}$, and the number of He atoms in the first solvation shell, calculated by integrating the helium density from the I position up to the first density profile minimum, is 20. Note also that two more shells are distinguishable. However, the concept of shell might not be particularly appropriate, since the density between the peaks of the second and the third shells is rather high.

3. $\text{I}^+@He_{2000}$

Buchachenko et al. have also calculated the He- I^+ 3Π and $3\Sigma^-$ pair potentials.¹³ Using them, we have addressed the structure of $\text{I}^+@He_{2000}$ when the cation is in the $X2$ (3Π state) (first Eq. (9) of Ref. 13). Due to the strong He- I^+ interaction, the equilibrium position of I^+ is at the center of the droplet.

The spherically averaged helium density profile of $\text{I}^+@^4\text{He}_{2000}$ at equilibrium, obtained when I^+ is in the 3Π state, is shown in Fig. 10. It can be seen that the effect of the impurity on the He droplet, even if it is sizeable, is local. The solvation shell of the I^+ cation is $S_{I^+} = -3578.1 \text{ K}$, and the number of atoms in the first solvation shell is 21.

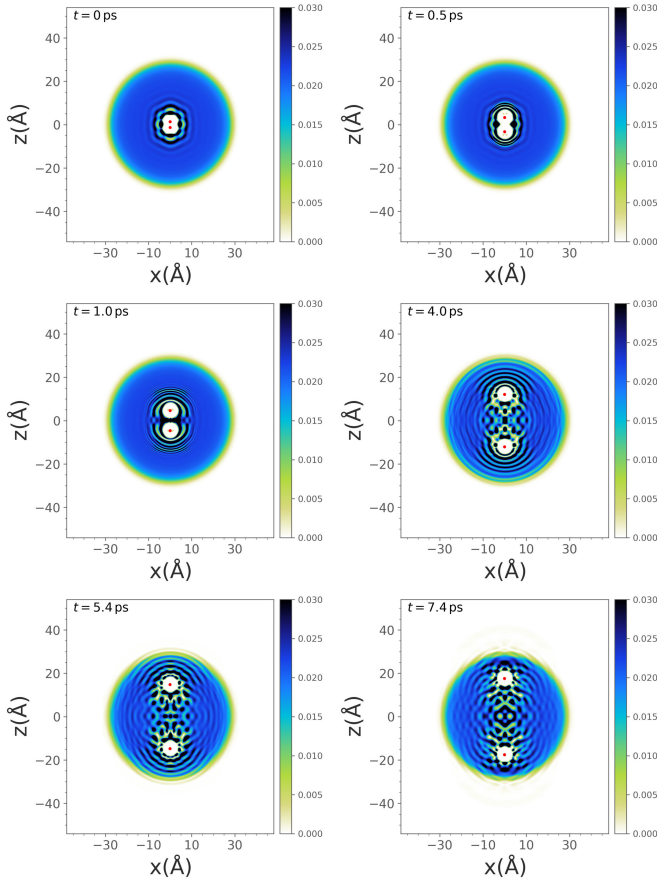


FIG. 11. Two-dimensional density plots of $I_2@He_{2000}$ with I_2 in the A state at different times starting from the molecule at $\mathbf{r} = 0$. The color bar represents the helium density in $\text{\AA}^{-3}$.

B. Dynamics

Starting from $I_2@^4He_{2000}$ in the ground state, the molecule is excited to the A state and the I atoms and He density are let to evolve with time following Eqs. (9) for a period of time to be determined. Fig. 11 shows one such evolution starting from $\mathbf{r} = 0$. Figure 12 shows the location of the two I atoms as a function of time, Fig. 13 the I-I distance as a function of time, and Fig. 14 shows the velocity of the I atoms (1 $\text{\AA}/\text{ps}$ corresponds to 100 m/s). They correspond to I_2 being originally at $\mathbf{r} = 0$. The same procedure can be followed when the I_2 is initially at any other location and orientation inside the droplet, as those shown e.g. in Fig. 7.

Fig. 15 shows the evolution when I_2 is at 24 $\text{\AA}$ from the droplet center radially oriented. The location of the I atoms of the I_2 molecule in the A state as a function of time is shown in Fig. 16, distance between the I atoms of the I_2 molecule in the A state as a function of time is shown in Fig. 17, and the velocity of the I atoms of the I_2 molecule in the A state as a function of time is shown in Fig. 18.

Fig. 19 shows the evolution when I_2 is at 24 $\text{\AA}$ from

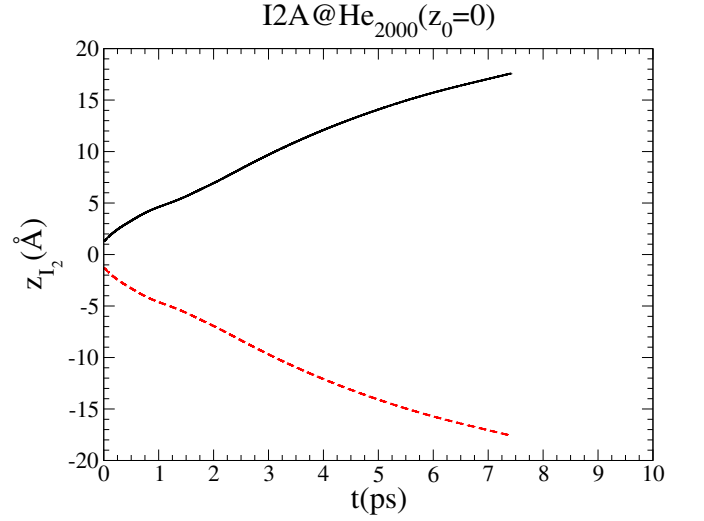


FIG. 12. Location of the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $I_2@He_{2000}$ at $\mathbf{r} = 0$

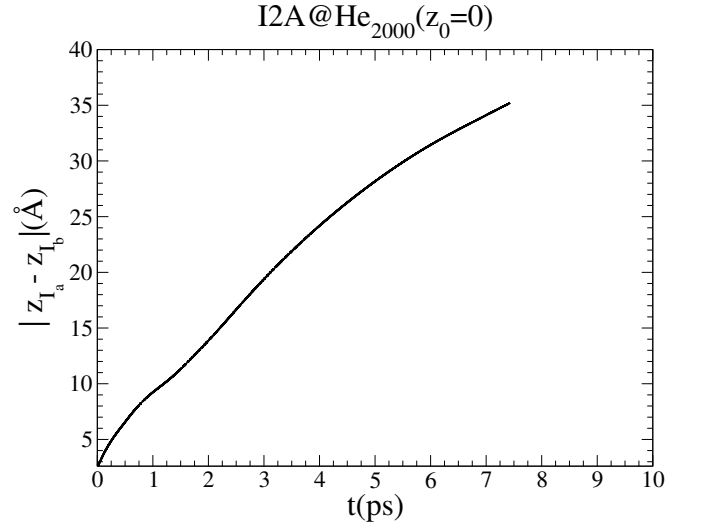


FIG. 13. Distance between the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $I_2@He_{2000}$ at $\mathbf{r} = 0$

the droplet center tangentially oriented. The location of the I atoms of the I_2 molecule in the A state as a function of time is shown in Fig. 20, distance between the I atoms of the I_2 molecule in the A state as a function of time is shown in Fig. 21, and the velocity of the I atoms of the I_2 molecule in the A state as a function of time is shown in Fig. 22.

After the I atoms of the molecule in the A state have evolved in time, the I atoms will be ionized. In practice, this is done by substituting I atoms by I^+ ions interacting with the He droplet through the $^3\Pi I^+-He$ pair potential¹³ **TO BE CONFIRMED** and among themselves with *only* the Coulomb force corresponding to point charges.

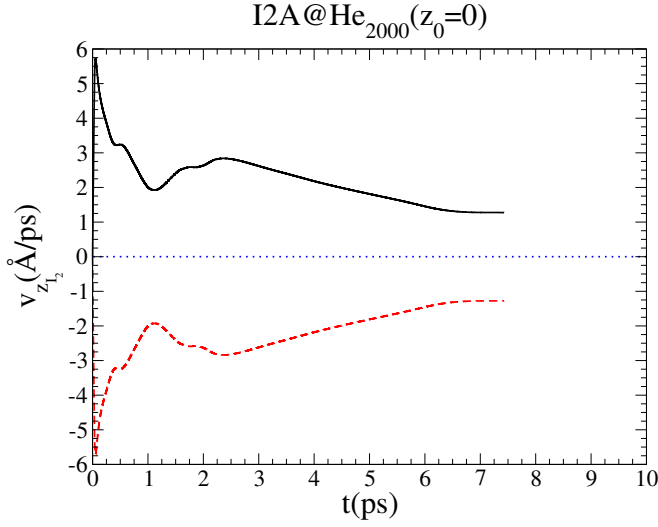


FIG. 14. Velocity of the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $I_2@He_{2000}$ at $\mathbf{r} = 0$

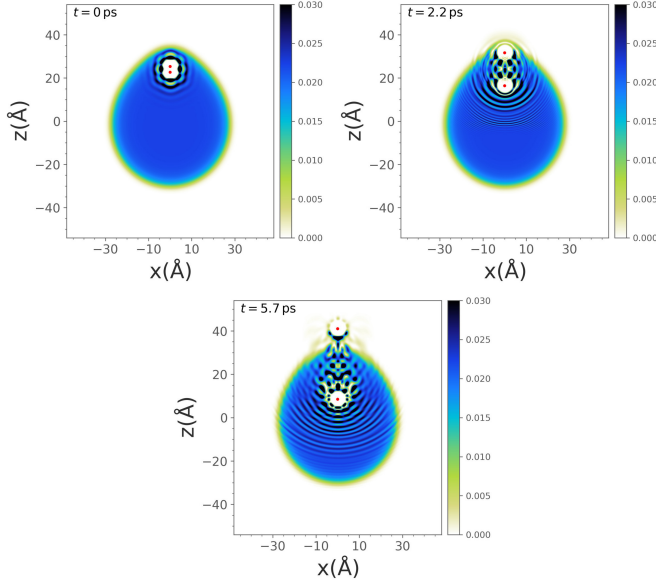


FIG. 15. Two-dimensional density plots of $I_2@He_{2000}$ with I_2 in the A state at different times. The starting configuration is $I_2@He_{2000}$ at $24 \text{ \AA}$ from the center of the droplet radially oriented. The color bar represents the helium density in $\text{\AA}^{-3}$. **TO be completed with other PANELS**

A word of caution: It is not obvious that simulating the excitation of the I_2 molecule to the A state yields the ejection of the I atoms, bare or with some helium atoms attached to them. We have had a situation like this in the dynamics of photoexcited Ba^+ cations,²³ where at variance with the experiments,²³ the simulations did not yield desorption of excited Ba^+ cations or exciplexes after the $6p \leftarrow 6s$ excitation. This “paradox” was solved in a subsequent work on the dynamics of $RbHe$ exciplexes off

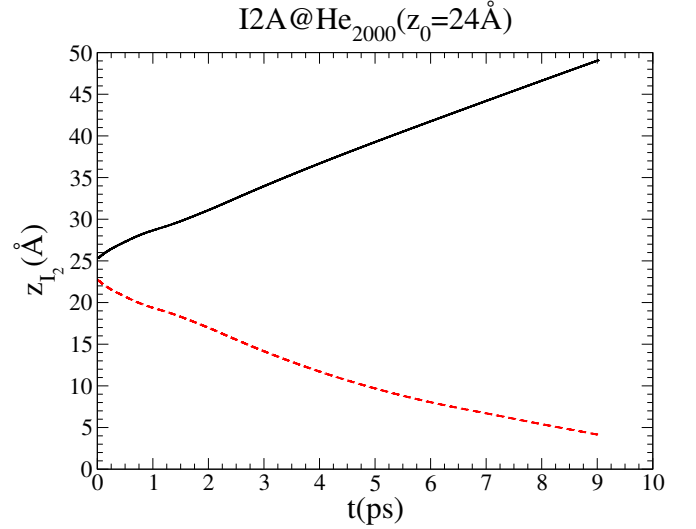


FIG. 16. Location of the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $I_2@He_{2000}$ at $24 \text{ \AA}$ from the center of the droplet radially oriented.

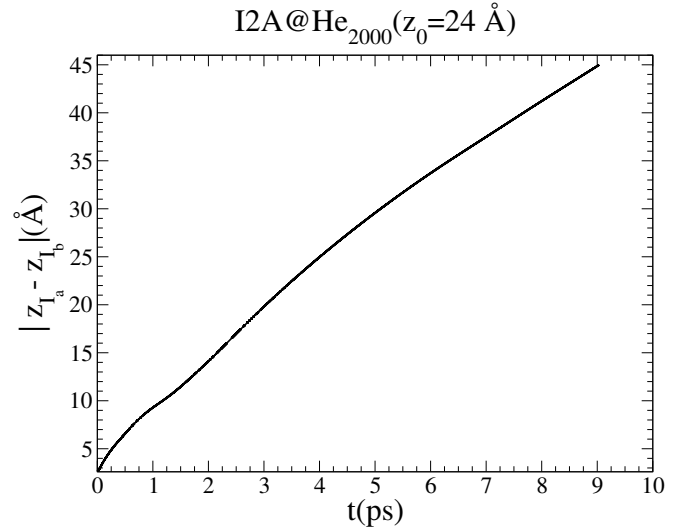


FIG. 17. Distance between the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $I_2@He_{2000}$ at $24 \text{ \AA}$ from the center of the droplet radially oriented.

helium nanodroplets,²⁵ where desorption of the exciplex was driven by its spin-relaxation.

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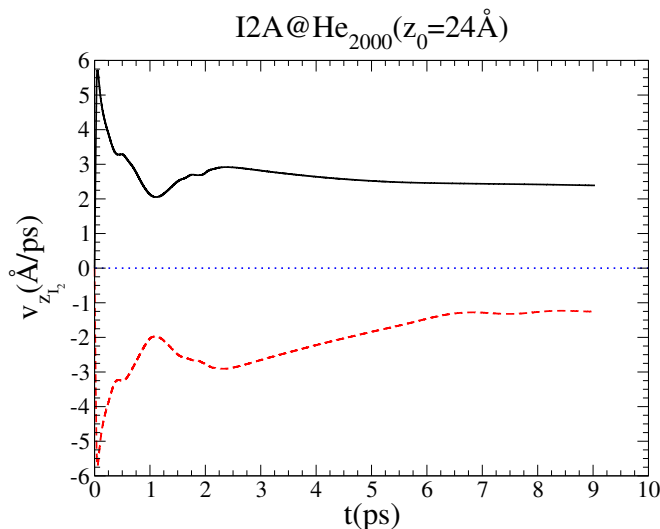


FIG. 18. Velocity of the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $I_2@He_{2000}$ at at 24 Å from the center of the droplet radially oriented.

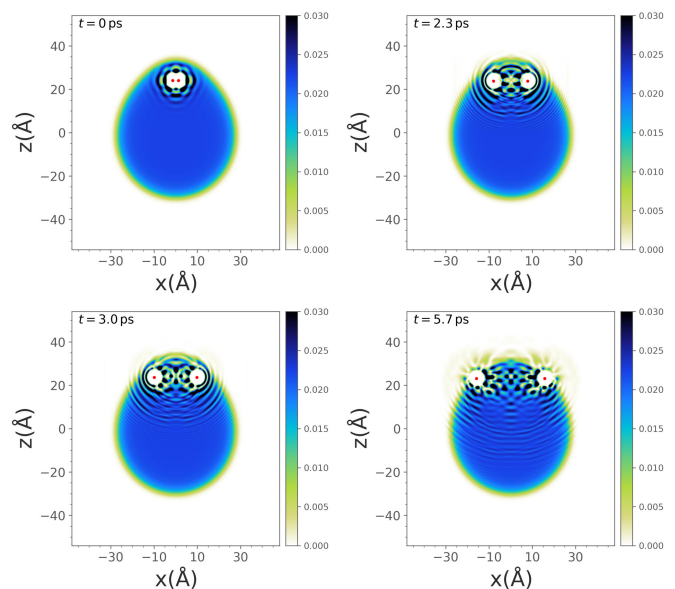


FIG. 19. Two-dimensional density plots of $I_2@He_{2000}$ with the I_2 molecule tangentially oriented at a distance of 24 Å from the center of the droplet. The color bar shows the density in units of $\text{\AA}^{-3}$. **TO be completed with other PANNELS**

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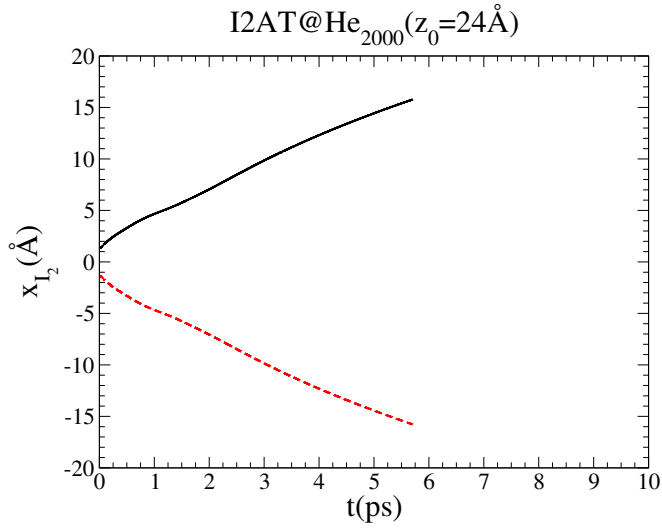


FIG. 20. Location of the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $I_2@He_{2000}$ at 24 Å from the center of the droplet tangentially oriented.

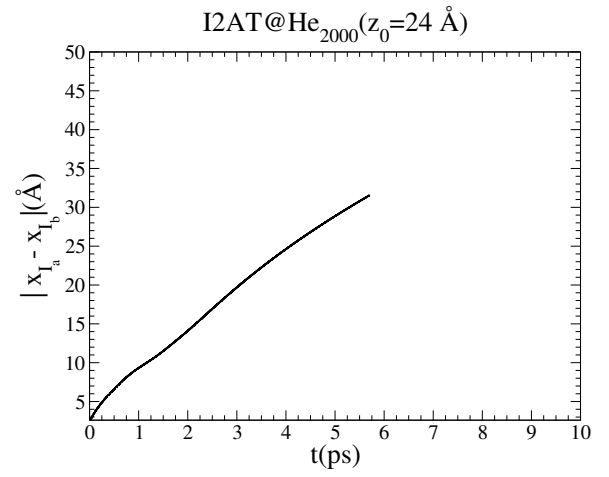


FIG. 21. Distance between the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $I_2@He_{2000}$ at 24 Å from the center of the droplet tangentially oriented.

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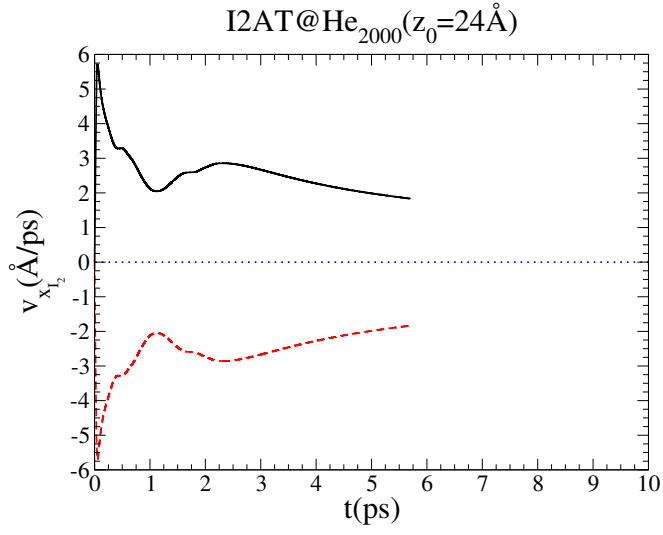


FIG. 22. Velocity of the I atoms of the I_2 molecule in the A state as a function of time. The starting configuration is $\text{I}_2\text{@He}_{2000}$ at at 24 Å from the center of the droplet tangentially oriented.